IntegralityFocus and the complete scaling workflow

Auxiliary OMS investigation

16 October 2026: aggregation clarification

Finding. On the known instance046 with presolve off, IF1 and IF0 return identical points, objectives, bounds and work in all six matched comparisons. All returned integer coordinates are exactly integral. L2 retains a 4,837.338 m^3 activation violation; Budget-L2 limits it to $1.4782 \times 10^{-5}\,\mathrm{m}^3$. This is a pertinent control, not evidence that IF1 is generally ineffective.

Design. Two policies, two IF settings and seeds 11, 23, 47: twelve principal calls. Each fresh process generates the original C++ model once, selects factors, exports, imports, solves and verifies. All Base/scaled exports reproduce the historical SHA-256 values. Gurobi 13.0.2 uses four threads, 120 seconds, 1% gap, presolve off, automatic scaling, primal tolerance 10^{-6} and integer tolerance 2^{-28}. No MIP starts, factor changes after observation or new reference LPs.

Accuracy and native reports. Demanding study targets are 0.01 MW and 100 m^3, not operator standards. Budgets counts original-unit checks after integer rounding. $B > U$ compares a reported lower bound with an exactly feasible objective of the same imported file, allowing $\max(10^{-6}, 8\,\mathrm{ulp}(B))$. Gap flags contradict a native 1% completion when $U < C - 0.01|C|$. Unflagged reports do not certify optimality. Work is the median of all three calls.

Policy	Calls	Budgets	Max. v_V (m^3)	$B > U$	Gap flags	Work
L2-window, IF0	3	0/3	4,837.338	3	3	0.597
L2-window, IF1	3	0/3	4,837.338	3	3	0.597
Budget-L2, IF0	3	3/3	1.478×10^{-5}	2	0	4.975
Budget-L2, IF1	3	3/3	1.478×10^{-5}	2	0	4.975

Budget retains two contradicted bounds of three despite no flagged completion at the requested 1% gap. Native termination and primal accuracy are separate.

Why 4.485 and 4.975 agree. The Budget-L2 Work values for seeds 11/23/47 are 3.6054776613, 4.9754950464 and 5.0281587478, identical to the earlier presolve-off campaign. Their geometric mean is 4.4847271866 (4.485 in manuscript historical Table 11 of v1.3.8 and Table S12), while their median is 4.9754950464 (4.975 here and in Table S15). Different summaries explain the figures; no data or settings changed.

Complete measured cost. Medians in seconds, three calls per cell. Generate includes C++ launch, input reading, construction and Base serialization. Select/export is the factor and scaled-file stage; import/check includes native environment setup and import equality checks. Total is the full fresh Python process, including startup, model reading, metadata, integrity, extraction and persistence. Additional stages are in the CSV; displayed stages are not an exhaustive partition and medians need not add.

Policy	Generate	Select/export	Import/check	Solve	Verify	Total
L2-window, IF0	0.115	0.428	0.114	0.639	0.269	2.070
L2-window, IF1	0.115	0.367	0.099	0.533	0.228	1.768
Budget-L2, IF0	0.065	0.514	0.097	4.788	0.237	6.231
Budget-L2, IF1	0.064	0.528	0.099	4.955	0.239	6.380

The machine was shared with another thesis computation. These measurements describe this prototype and do not establish a general speed comparison at equal quality. Generator compilation and previous witness acquisition are offline costs.

Instrumentation amendment. The first twelve-call pass is retained in `first-pass/`. It repeatedly copied binding arrays inside its import checker. After caching those arrays once, all twelve treatments were repeated with unchanged models, factors, seeds, parameters and checks. All saved points and native outcomes match exactly across passes. There are 24 new MILP calls in total; the first pass is not pooled as an independent sample or used for the principal timing table.

Independent replay. A separate rational checker revalidates both exact feasible witnesses (20,160 rows, 12,240 variables, 2,880 integer domains) and checks 103,680 physical row/point combinations before/after rounding. It uses saved data and an open-source model reader, without optimization or a commercial license. Four endpoint tests cover rounding, exact cancellation and bound/gap refutations. Witness revalidation cost is separate from the workflow. Neither witness is a global MILP optimum certificate.

OMS decision. Add a short matched-control statement in Section 7.9; retain detail and timing in the supplement. Preserve the two contributions, abstract and introduction; the v1.3.8 editorial closure also shortens the cover letter. This closes a specific comparison rather than opening a new mechanism campaign.